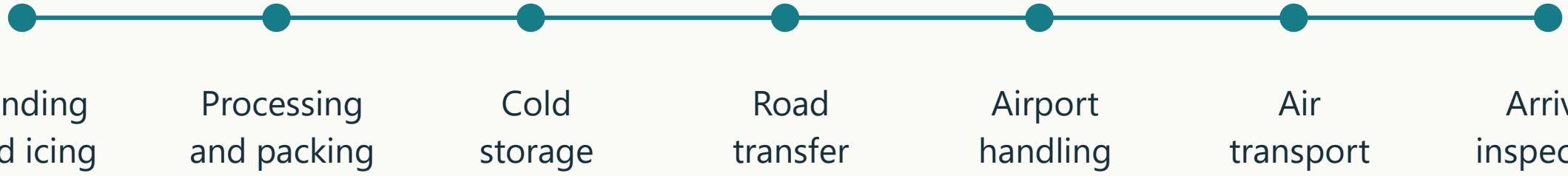
Digital Twin of Indian Fisheries Export Cold Chain for Spoilage Risk Prediction

Testing where a chilled-fish export journey picks up heat.

RESEARCH PROTOTYPE · INDIA → EUROPE · OCTOBER 2026

Fish passes through many hands.
The cold chain must stay connected.

An illustrative export journey from India to Europe



Each handover can change the time spent in warm conditions.

Small disruptions can add heat.

Temperature and time both matter across the journey.

Cooling stops

A refrigerated vehicle loses cooling.

Handling takes longer

Fish waits during loading or inspection.

Doors stay open

Warm air enters a chilled space.

Setpoints rise

Storage or transport runs warmer.

Where does a disruption add the most temperature exposure to an export journey?

LONGER-TERM AIM

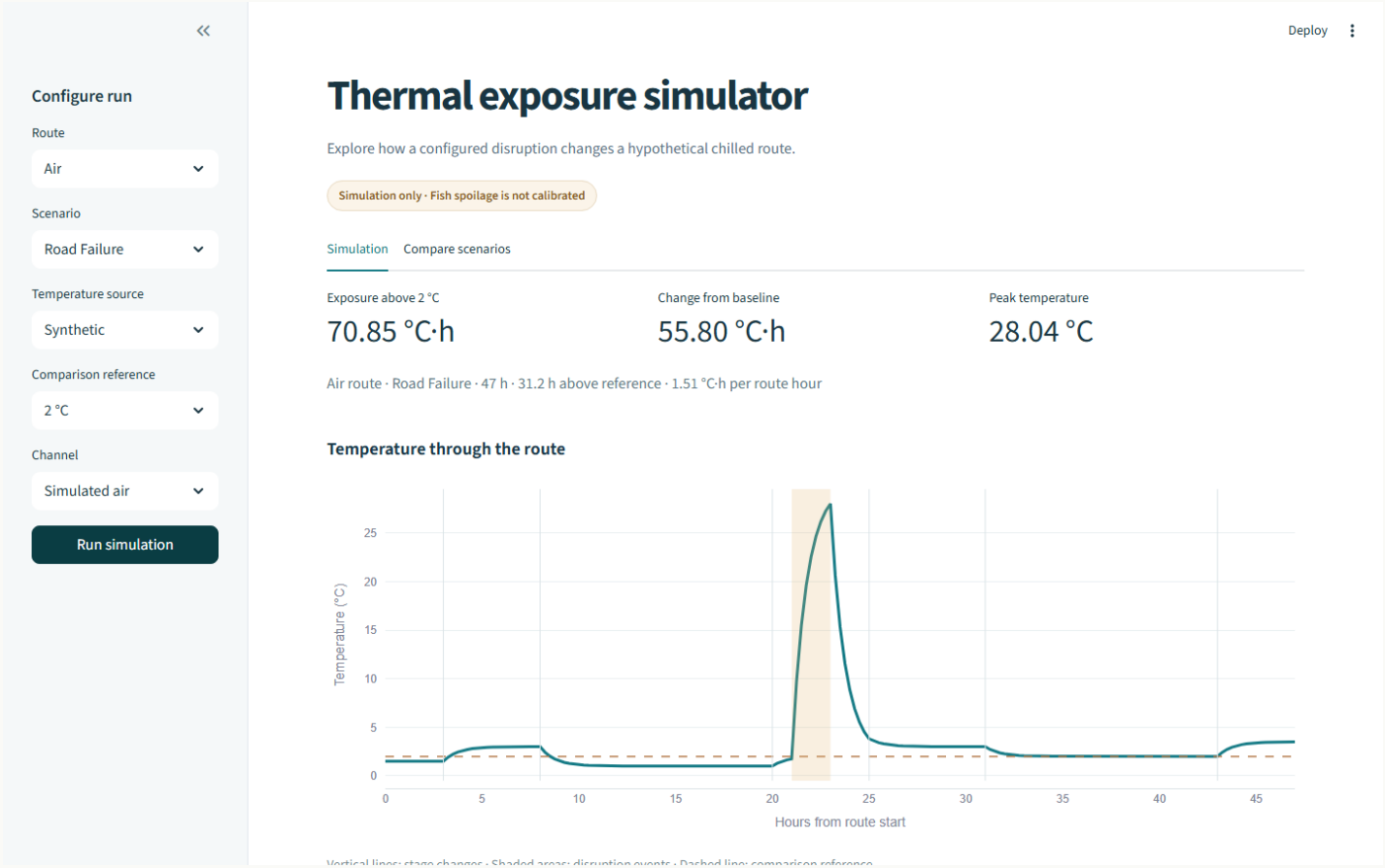
Use paired fish-quality measurements to turn exposure into a tested spoilage forecast.

A virtual cold chain we can experiment with.

Change the cooling. See the effect.

A computer model follows the journey and compares normal cooling with disruptions.

Working local dashboard



Real measurements, used within their limits.

The sources play different roles in the system.

Fish sensors	DaFiF and rainbow-trout sensor readings	Study fish measurements; no matched quality labels
Trout quality	21 measured microbial-count assays	A small, separate study at nominal 4 °C
Export statistics	Official MPEDA records	Indian trade context; not shipment tracking
Thermal records	Citrus reefer and cold-room measurements	Nonfish proxy patterns; not seafood route logs

From a route choice to a stage comparison.

01 **Choose a route** → 02 **Set normal cooling** → 03 **Introduce a disruption** → 04 **Calculate exposure** → 05 **Compare the stages**

Example: temperature stays at 5 °C for 2 hours

3 °C Above 2 °C reference × **2 h** Duration = **6 °C·h** Thermal exposure

This is a comparison measure, not a universal spoilage threshold.

What if the journey does not go to plan?

Each disruption is compared with the same route under normal cooling.

Normal cooling

The reference journey

Road cooling failure

Cooling stops for 2 hours

Airport delay

Extra waiting and an open door

Warmer storage

Setpoint rises by 3 °C

Flight excursion

Setpoint rises for 4 hours

Combined failures

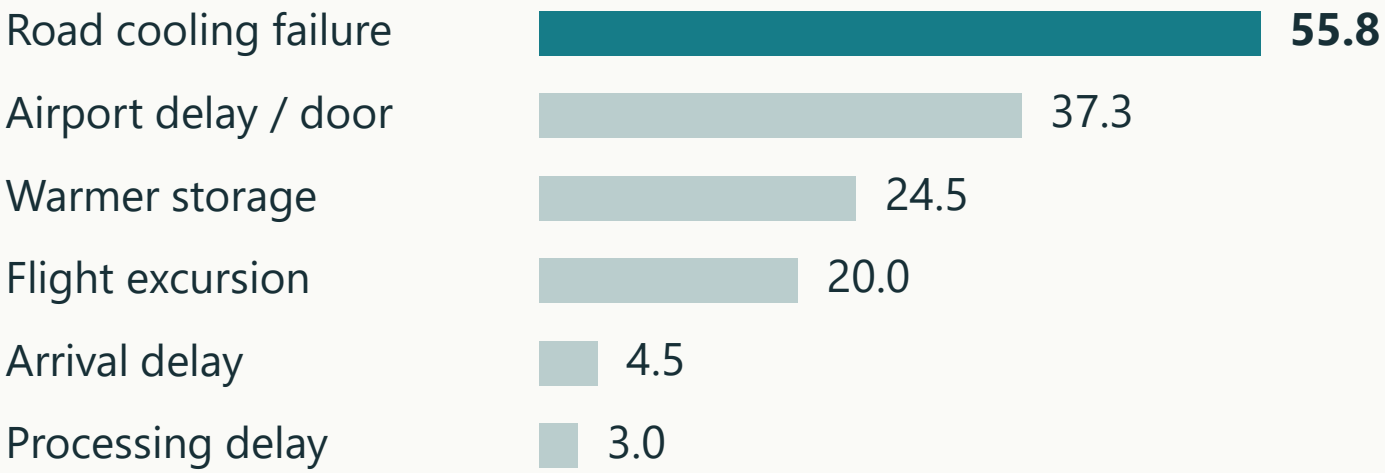
Road, airport and flight together

Processing and arrival delays were also tested. All routes and disruptions are hypothetical.

Road cooling failure added the most exposure.

Among the configured single-disruption air-route scenarios

ADDED EXPOSURE VS NORMAL COOLING · °C·h ABOVE 2 °C



55.8

additional °C·h

from a 2-hour
road cooling failure

Simulated air · Hypothetical 47 h route · Disruption sizes differ, so this is a conditional ranking.

The road result held across most tested assumptions.

197 / 200

When we changed our simulation assumptions 200 times, road cooling failure was the biggest contributor in 197 cases.

Air route · 2 °C comparison reference

The ranking is not universal.

It changed when we used equal-sized disruptions or changed the comparison temperature.

These are hypothetical assumption tests, not validation against real fish shipments.

We can compare heat exposure. Spoilage prediction needs more evidence.

AVAILABLE NOW

Simulate a configured journey
Compare disruption effects
Inspect and download results

NEEDED FOR SPOILAGE PREDICTION

Fish temperature and quality measured together
Independent shipment tests

The planned spoilage classifier has not been trained: specimen-matched labels are missing.

We built a working virtual cold chain.

It identifies where simulated temperature exposure increases.

The next step is connecting those simulations with real fish-quality measurements to predict spoilage.

01

Measure

Collect paired fish-temperature and quality data.

02

Validate

Test on independent Indian export shipments.

03

Prepare research paper

Document findings and evidence limits.